HW2_3_course

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Individual project: Exploratory factor analysis

```
library(foreign)
library(corrplot)
library(ggplot2)
library(lavaan)
library(tidyr)
library(psych)
library(dplyr)
library(nFactors)
library(knitr)
library(kableExtra)
```

1. Filtering data and removing NAs

```
ess <- read.csv('/home/lena/Загрузки/ESS2e03_6.csv', sep = ',')

estonia <- ess %>%
  filter(cntry == "EE")

est <- estonia %>%
  select(ipcrtiv, imprich, ipeqopt, ipshabt, impsafe, impdiff,
         ipfrule, ipudrst, ipmodst, ipgdtim, impfree, iphlppl,
         ipsuces, ipstrgv, ipadvnt, ipbhprp, iprspot, iplylfr,
         impenv, imptrad, impfun) %>%
  mutate(across(everything(), ~ifelse(. %in% c(7, 8, 9), NA, .))) %>%
  mutate(across(everything(), as.numeric)) %>%
  drop_na()

sapply(est, class)
```

```
##   ipcrtiv   imprich   ipeqopt   ipshabt   impsafe   impdiff   ipfru
le   ipudrst
## "numeric" "numeric" "numeric" "numeric" "numeric" "numeric" "numeri
c" "numeric"
##   ipmodst   ipgdtim   impfree   iphlppl   ipsuces   ipstrgv   ipadv
nt   ipbhprp
## "numeric" "numeric" "numeric" "numeric" "numeric" "numeric" "numeri
c" "numeric"
##   iprspot   iplylfr    impenv   imptrad    impfun
## "numeric" "numeric" "numeric" "numeric" "numeric"
```

2. Description of variables

```
variable_names <- c("ipcrtiv", "imprich", "ipeqopt", "ipshabt", "impsa
fe",
                    "impdiff", "ipfrule", "ipudrst", "ipmodst", "ipgdt
im",
                    "impfree", "iphlppl", "ipsuces", "ipstrgv", "ipadv
nt",
                    "ipbhprp", "iprspot", "iplylfr", "impenv", "imptra
d",
                    "impfun")

meanings <- c("Important to think new ideas and being creative",
              "Important to be rich, have money and expensive things",
              "Important that people are treated equally and have equa
l opportunities",
              "Important to show abilities and be admired",
              "Important to live in secure and safe surroundings",
              "Important to try new and different things in life",
              "Important to do what is told and follow rules",
              "Important to understand different people",
              "Important to be humble and modest, not draw attention",
              "Important to have a good time",
              "Important to make own decisions and be free",
              "Important to help people and care for others well-bein
g",
              "Important to be successful and that people recognize ac
hievements",
              "Important that government is strong and ensures safet
y",
              "Important to seek adventures and have an exciting lif
e",
              "Important to behave properly",
              "Important to get respect from others",
              "Important to be loyal to friends and devote to people c
lose",
              "Important to care for nature and environment",
              "Important to follow traditions and customs",
              "Important to seek fun and things that give pleasure")

scales <- rep("1 - Very much like me, 6 - Not like me at all", 21)

tab <- data.frame(
  Variable = variable_names,
  Meaning = meanings,
  Scale = scales,
  stringsAsFactors = FALSE
)
```

```
kable(tab) %>%
  kable_styling(bootstrap_options = c("bordered", "responsive", "strip
ed"), full_width = FALSE)
```

Variable	Meaning	Scale
ipcrtiv	Important to think new ideas and being creative	1 - Very much like me, 6 - Not like me at all
imprich	Important to be rich, have money and expensive things	1 - Very much like me, 6 - Not like me at all
ipeqopt	Important that people are treated equally and have equal opportunities	1 - Very much like me, 6 - Not like me at all
ipshabt	Important to show abilities and be admired	1 - Very much like me, 6 - Not like me at all
impsafe	Important to live in secure and safe surroundings	1 - Very much like me, 6 - Not like me at all
impdiff	Important to try new and different things in life	1 - Very much like me, 6 - Not like me at all
ipfrule	Important to do what is told and follow rules	1 - Very much like me, 6 - Not like me at all
ipudrst	Important to understand different people	1 - Very much like me, 6 - Not like me at all
ipmodst	Important to be humble and modest, not draw attention	1 - Very much like me, 6 - Not like me at all
ipgdtim	Important to have a good time	1 - Very much like me, 6 - Not like me at all
impfree	Important to make own decisions and be free	1 - Very much like me, 6 - Not like me at all
iphlppl	Important to help people and care for others well-being	1 - Very much like me, 6 - Not like me at all
ipsuces	Important to be successful and that people recognize achievements	1 - Very much like me, 6 - Not like me at all
ipstrgv	Important that government is strong and ensures safety	1 - Very much like me, 6 - Not like me at all

Variable	Meaning	Scale
ipadvnt	Important to seek adventures and have an exciting life	1 - Very much like me, 6 - Not like me at all
ipbhprp	Important to behave properly	1 - Very much like me, 6 - Not like me at all
iprspot	Important to get respect from others	1 - Very much like me, 6 - Not like me at all
iplylfr	Important to be loyal to friends and devote to people close	1 - Very much like me, 6 - Not like me at all
impenv	Important to care for nature and environment	1 - Very much like me, 6 - Not like me at all
imptrad	Important to follow traditions and customs	1 - Very much like me, 6 - Not like me at all
impfun	Important to seek fun and things that give pleasure	1 - Very much like me, 6 - Not like me at all

3. Checking assumtions for FA

3.1 Sample size

```
n_cases <- nrow(est)
n_vars <- ncol(est)
cat("Sample size:", n_cases, "\n")
```

```
## Sample size: 1795
```

```
cat("Number of variables:", n_vars, "\n")
```

```
## Number of variables: 21
```

```
cat("Cases per variable:", round(n_cases/n_vars, 1), "\n")
```

```
## Cases per variable: 85.5
```

3.2 Normality

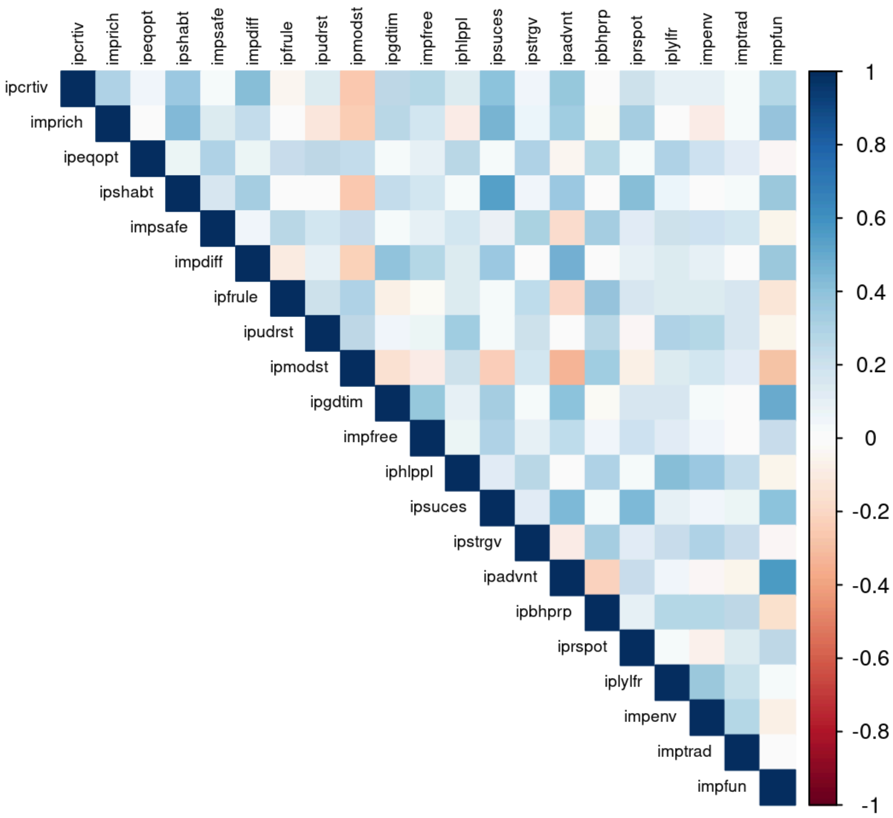
```
describe(est)
```

```
##         vars    n mean   sd median trimmed  mad min max range  skew
kurtosis
## ipcrtiv    1 1795 3.11 1.27      3    3.06 1.48   1   6     5  0.38
-0.56
## imprich    2 1795 3.94 1.26      4    3.97 1.48   1   6     5 -0.25
-0.89
## ipeqopt    3 1795 2.40 1.08      2    2.28 1.48   1   6     5  0.99
0.70
## ipshabt    4 1795 3.73 1.35      4    3.73 1.48   1   6     5 -0.12
-1.02
## impsafe    5 1795 2.39 1.06      2    2.28 0.00   1   6     5  1.07
1.01
## impdiff    6 1795 3.16 1.30      3    3.12 1.48   1   6     5  0.32
-0.83
## ipfrule    7 1795 3.27 1.30      3    3.23 1.48   1   6     5  0.27
-0.93
## ipudrst    8 1795 2.48 0.96      2    2.41 0.00   1   6     5  0.92
0.89
## ipmodst    9 1795 2.95 1.24      3    2.90 1.48   1   6     5  0.45
-0.60
## ipgdtim   10 1795 3.00 1.25      3    2.94 1.48   1   6     5  0.48
-0.61
## impfree   11 1795 2.38 1.03      2    2.29 1.48   1   6     5  0.92
0.73
## iphlppl   12 1795 2.58 0.94      2    2.55 1.48   1   6     5  0.58
0.41
## ipsuces   13 1795 3.45 1.27      3    3.43 1.48   1   6     5  0.11
-0.91
## ipstrgv   14 1795 2.30 1.02      2    2.17 0.00   1   6     5  1.18
1.58
## ipadvnt   15 1795 4.13 1.36      4    4.21 1.48   1   6     5 -0.48
-0.71
## ipbhprp   16 1795 2.68 1.12      2    2.61 1.48   1   6     5  0.73
0.01
## iprspot   17 1795 3.91 1.31      4    3.93 1.48   1   6     5 -0.24
-0.97
## iplylfr   18 1795 2.10 0.79      2    2.05 0.00   1   6     5  0.97
2.00
## impenv    19 1795 2.04 0.87      2    1.96 0.00   1   6     5  0.98
1.49
## imptrad   20 1795 3.12 1.32      3    3.08 1.48   1   6     5  0.37
-0.75
## impfun    21 1795 3.91 1.37      4    3.93 1.48   1   6     5 -0.20
-0.93
##           se
## ipcrtiv 0.03
## imprich 0.03
## ipeqopt 0.03
## ipshabt 0.03
```

```
## impsafe 0.03
## impdiff 0.03
## ipfrule 0.03
## ipudrst 0.02
## ipmodst 0.03
## ipgdtim 0.03
## impfree 0.02
## iphlppl 0.02
## ipsuces 0.03
## ipstrgv 0.02
## ipadvnt 0.03
## ipbhprp 0.03
## iprspot 0.03
## iplylfr 0.02
## impenv  0.02
## imptrad 0.03
## impfun  0.03
```

3.3 Linearity and multicollinearity

```
cor_matrix <- cor(est, use = "complete.obs")
corrplot(cor_matrix, method = "color", type = "upper",
         tl.cex = 0.6, tl.col = "black")
```




```
det(cor_matrix)
```

```
## [1] 0.003976691
```

3.4 Sampling adequacy

```
KMO(est)
```

```
## Kaiser-Meyer-Olkin factor adequacy
## Call: KMO(r = est)
## Overall MSA =  0.86
## MSA for each item =
## ipcrtiv imprich ipeqopt ipshabt impsafe impdiff ipfrule ipudrst ipm
odst ipgdtim
##    0.88    0.89    0.86    0.87    0.83    0.87    0.85    0.85
0.88    0.84
## impfree iphlppl ipsuces ipstrgv ipadvnt ipbhprp iprspot iplylfr  im
penv imptrad
##    0.86    0.85    0.88    0.88    0.87    0.88    0.81    0.86
0.85    0.85
##  impfun
##    0.87
```

3.5 Sphericity

```
cortest.bartlett(est)
```

```
## R was not square, finding R from data
```

```
## $chisq
## [1] 9872.688
##
## $p.value
## [1] 0
##
## $df
## [1] 210
```

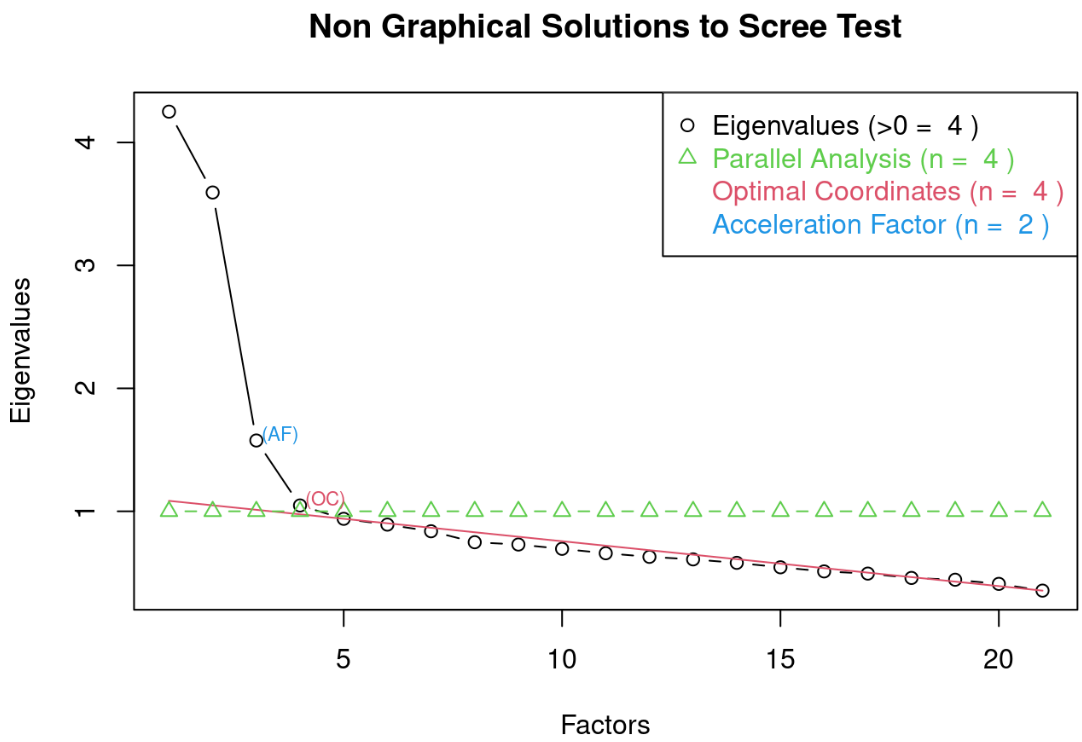
Sample size and cases per variable are good. Variables are distributed within acceptable range (skewness and kurtosis are within ±2 range). Correlation matrix shows that there are linear relationships between some of the variables which is enough for FA. det = 0.00035 shows normal multicollinearity which means that connections are strong enough for FA. MSA for each variable > 0.8 which means that variables group well. Bartlett test shows that variables are connected (p < 0.05). All assumptions have been met.

4. Determinig the number of factors

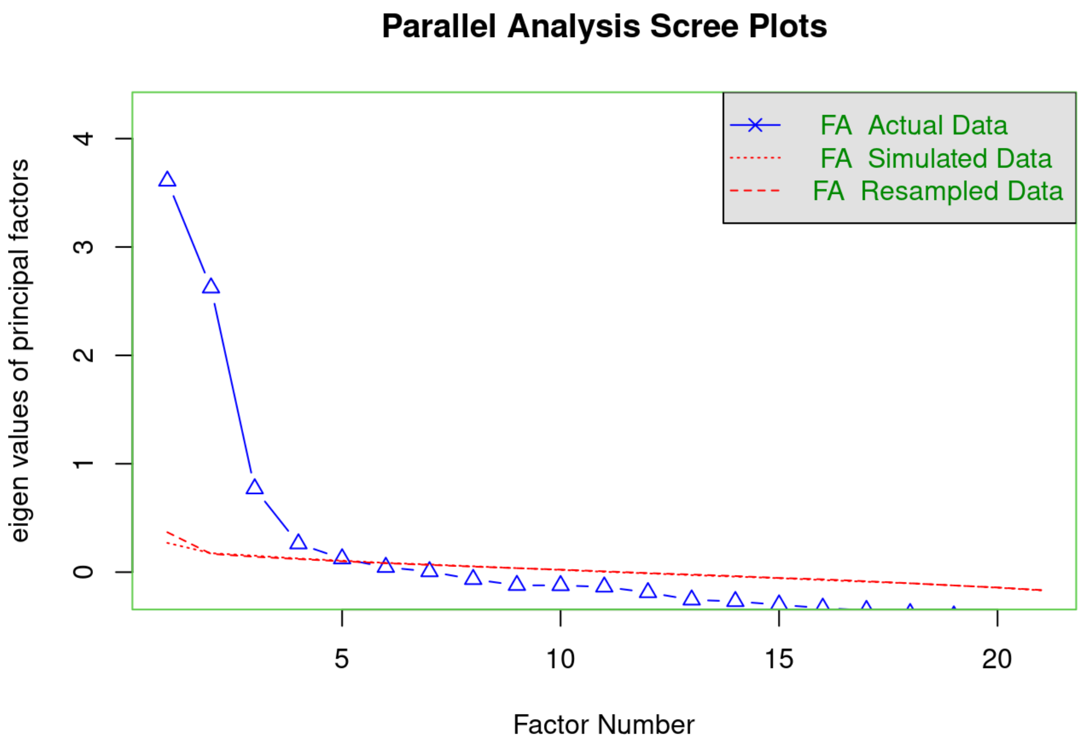
```
ev <- eigen(cor(est))
ev$values
```

```
##  [1] 4.2509646 3.5932790 1.5759077 1.0474071 0.9388261 0.8916045 0.
8375669
##  [8] 0.7483126 0.7297168 0.6946515 0.6586820 0.6295134 0.6087827 0.
5807315
## [15] 0.5444672 0.5109194 0.4932874 0.4578522 0.4432755 0.4095270 0.
3547250
```

```
nS <- nScree(x=ev$values, model = 'factors')
plotnScree(nS)
```




```
fa.parallel(est, fm = 'ml', fa = 'fa')
```




```
## Parallel analysis suggests that the number of factors =  5  and the
number of components =  NA
```

Eigenvalues analysis shows 4 factors containig enough information (>1). Scree plot also shows 4 as an optimal number of factors and parallel analysis suggests that the number of factors should be 5. I will try to conduct the analysis with 4 and 5 factors.

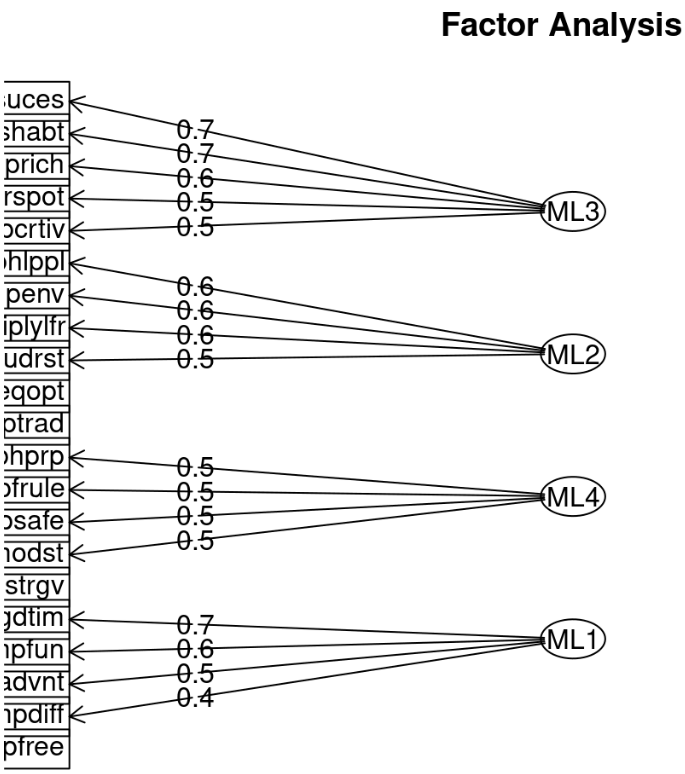
5. Conducting EFA

```
fa1 = fa(est, nfactors = 4,fm = 'ml', rotate = 'varimax')
fa1
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 4, rotate = "varimax", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML3   ML2   ML4   ML1   h2   u2 com
## ipcrtiv  0.48  0.26 -0.18  0.22 0.38 0.62 2.3
## imprich  0.57 -0.15  0.09  0.26 0.42 0.58 1.6
## ipeqopt  0.02  0.36  0.32  0.05 0.24 0.76 2.0
## ipshabt  0.70  0.02  0.04  0.16 0.51 0.49 1.1
## impsafe  0.12  0.22  0.49  0.00 0.31 0.69 1.5
## impdiff  0.35  0.27 -0.22  0.42 0.42 0.58 3.3
## ipfrule  0.02  0.15  0.52 -0.11 0.30 0.70 1.3
## ipudrst -0.05  0.51  0.16  0.05 0.29 0.71 1.2
## ipmodst -0.36  0.21  0.46 -0.12 0.40 0.60 2.5
## ipgdtim  0.17  0.08 -0.01  0.71 0.55 0.45 1.1
## impfree  0.23  0.13  0.04  0.37 0.21 0.79 2.0
## iphlppl  0.02  0.62  0.15  0.04 0.41 0.59 1.1
## ipsuces  0.71  0.10  0.04  0.26 0.58 0.42 1.3
## ipstrgv  0.09  0.33  0.40  0.01 0.28 0.72 2.1
## ipadvnt  0.44  0.03 -0.34  0.51 0.56 0.44 2.7
## ipbhprp -0.01  0.37  0.54 -0.06 0.43 0.57 1.8
## iprspot  0.55 -0.10  0.27  0.10 0.39 0.61 1.6
## iplylfr  0.01  0.55  0.18  0.16 0.36 0.64 1.4
## impenv  -0.03  0.56  0.13  0.00 0.33 0.67 1.1
## imptrad  0.07  0.29  0.25 -0.03 0.15 0.85 2.1
## impfun   0.38 -0.11 -0.13  0.61 0.55 0.45 1.9
##
##                        ML3  ML2  ML4  ML1
## SS loadings           2.55 2.05 1.74 1.74
## Proportion Var        0.12 0.10 0.08 0.08
## Cumulative Var        0.12 0.22 0.30 0.38
## Proportion Explained  0.32 0.25 0.22 0.22
## Cumulative Proportion 0.32 0.57 0.78 1.00
##
## Mean item complexity =  1.8
## Test of the hypothesis that 4 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 132  and the objective function was  0.38
##
## The root mean square of the residuals (RMSR) is  0.03
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  552.55
with prob <  8e-53
## The total n.obs was  1795  with Likelihood Chi Square =  669.79  wi
th prob <  7.1e-73
##
```

```
## Tucker Lewis Index of factoring reliability =  0.911
## RMSEA index =  0.048  and the 90 % confidence intervals are  0.044
0.051
## BIC =  -319.26
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML3  ML2  ML4  M
L1
## Correlation of (regression) scores with factors   0.87 0.84 0.82 0.
81
## Multiple R square of scores with factors          0.76 0.71 0.67 0.
66
## Minimum correlation of possible factor scores     0.51 0.41 0.33 0.
32
```

```
fa.diagram(fa1, cut = 0.4)
```




```
print(fa1$loadings,cutoff = 0.4)
```

```
##
## Loadings:
##         ML3    ML2    ML4    ML1
## ipcrtiv  0.485
## imprich  0.567
## ipeqopt
## ipshabt  0.698
## impsafe                0.491
## impdiff                       0.421
## ipfrule                0.520
## ipudrst         0.509
## ipmodst                0.461
## ipgdtim                       0.715
## impfree
## iphlppl         0.619
## ipsuces  0.709
## ipstrgv
## ipadvnt  0.436                0.509
## ipbhprp                0.540
## iprspot  0.549
## iplylfr         0.552
## impenv          0.561
## imptrad
## impfun                        0.611
##
##                  ML3   ML2   ML4   ML1
## SS loadings    2.549 2.052 1.740 1.737
## Proportion Var 0.121 0.098 0.083 0.083
## Cumulative Var 0.121 0.219 0.302 0.385
```

```
fa2 = fa(est, nfactors = 4, fm = 'ml',rotate = 'oblimin')
```

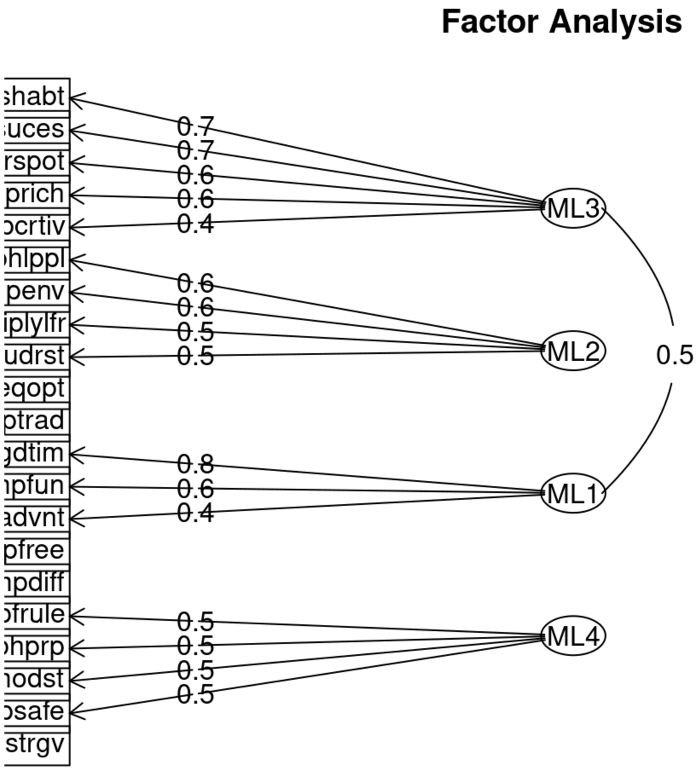
```
## Загрузка требуемого пакета: GPArotation
```

```
fa2
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 4, rotate = "oblimin", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML3   ML2   ML1   ML4   h2   u2 com
## ipcrtiv  0.41  0.30  0.05 -0.28 0.38 0.62 2.7
## imprich  0.55 -0.18  0.17  0.08 0.42 0.58 1.5
## ipeqopt  0.03  0.33  0.05  0.27 0.24 0.76 2.0
## ipshabt  0.72  0.02 -0.02 -0.04 0.51 0.49 1.0
## impsafe  0.20  0.16  0.00  0.46 0.31 0.69 1.6
## impdiff  0.18  0.30  0.33 -0.27 0.42 0.58 3.5
## ipfrule  0.14  0.08 -0.08  0.50 0.30 0.70 1.3
## ipudrst -0.07  0.51  0.02  0.09 0.29 0.71 1.1
## ipmodst -0.29  0.14  0.00  0.49 0.40 0.60 1.8
## ipgdtim -0.08  0.04  0.79  0.07 0.55 0.45 1.0
## impfree  0.11  0.11  0.37  0.05 0.21 0.79 1.4
## iphlppl  0.00  0.63 -0.03  0.05 0.41 0.59 1.0
## ipsuces  0.69  0.09  0.10 -0.04 0.58 0.42 1.1
## ipstrgv  0.15  0.29 -0.01  0.35 0.28 0.72 2.3
## ipadvnt  0.23  0.06  0.43 -0.34 0.56 0.44 2.6
## ipbhprp  0.08  0.30 -0.05  0.49 0.43 0.57 1.7
## iprspot  0.62 -0.14  0.01  0.24 0.39 0.61 1.4
## iplylfr -0.04  0.54  0.13  0.11 0.36 0.64 1.2
## impenv  -0.04  0.57 -0.05  0.04 0.33 0.67 1.0
## imptrad  0.12  0.27 -0.06  0.19 0.15 0.85 2.4
## impfun   0.18 -0.12  0.61 -0.07 0.55 0.45 1.3
##
##                        ML3  ML2  ML1  ML4
## SS loadings           2.39 2.12 1.80 1.76
## Proportion Var        0.11 0.10 0.09 0.08
## Cumulative Var        0.11 0.21 0.30 0.38
## Proportion Explained  0.30 0.26 0.22 0.22
## Cumulative Proportion 0.30 0.56 0.78 1.00
##
##  With factor correlations of
##       ML3  ML2   ML1   ML4
## ML3  1.00 0.10  0.53 -0.11
## ML2  0.10 1.00  0.15  0.28
## ML1  0.53 0.15  1.00 -0.29
## ML4 -0.11 0.28 -0.29  1.00
##
## Mean item complexity =  1.7
## Test of the hypothesis that 4 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 132  and the objective function was  0.38
##
## The root mean square of the residuals (RMSR) is  0.03
```

```
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  552.55
with prob <  8e-53
## The total n.obs was  1795  with Likelihood Chi Square =  669.79  wi
th prob <  7.1e-73
##
## Tucker Lewis Index of factoring reliability =  0.911
## RMSEA index =  0.048  and the 90 % confidence intervals are  0.044
0.051
## BIC =  -319.26
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML3  ML2  ML1  M
L4
## Correlation of (regression) scores with factors   0.91 0.87 0.89 0.
85
## Multiple R square of scores with factors          0.82 0.76 0.79 0.
73
## Minimum correlation of possible factor scores     0.64 0.52 0.57 0.
46
```

```
fa.diagram(fa2, cut = 0.4)
```




```
print(fa2$loadings,cutoff = 0.4)
```

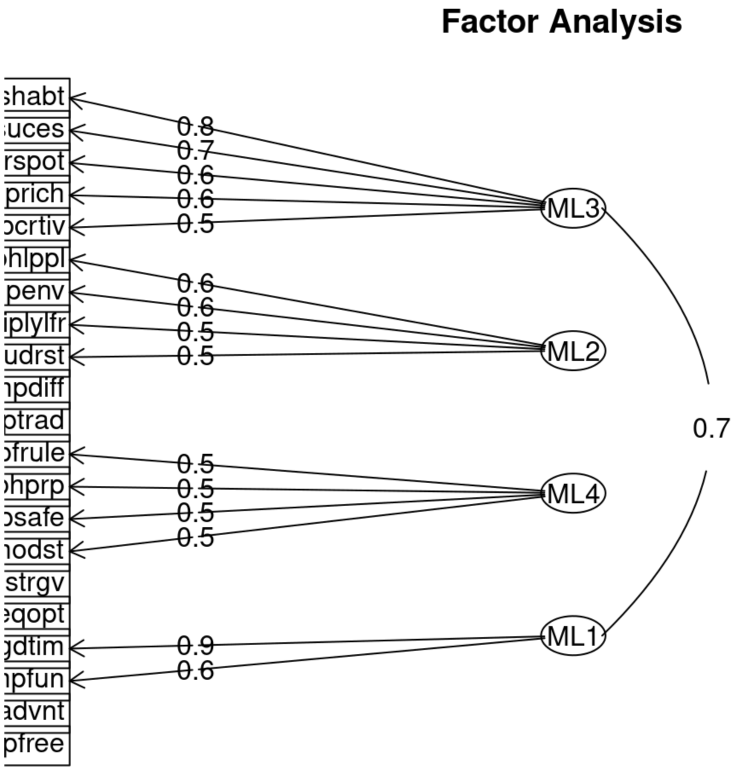
```
##
## Loadings:
##         ML3    ML2    ML1    ML4
## ipcrtiv  0.413
## imprich  0.551
## ipeqopt
## ipshabt  0.718
## impsafe                       0.460
## impdiff
## ipfrule                       0.503
## ipudrst         0.507
## ipmodst                       0.486
## ipgdtim                0.791
## impfree
## iphlppl         0.627
## ipsuces  0.686
## ipstrgv
## ipadvnt                0.427
## ipbhprp                       0.495
## iprspot  0.616
## iplylfr         0.544
## impenv          0.570
## imptrad
## impfun                 0.614
##
##                  ML3   ML2   ML1   ML4
## SS loadings    2.173 1.954 1.505 1.554
## Proportion Var 0.103 0.093 0.072 0.074
## Cumulative Var 0.103 0.197 0.268 0.342
```

```
fa3 = fa(est, nfactors = 4, fm = 'ml',rotate = 'promax')
fa3
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 4, rotate = "promax", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML3   ML2   ML4   ML1   h2   u2 com
## ipcrtiv  0.48  0.31 -0.23 -0.03 0.38 0.62 2.2
## imprich  0.57 -0.24  0.16  0.15 0.42 0.58 1.7
## ipeqopt -0.01  0.29  0.29  0.06 0.24 0.76 2.1
## ipshabt  0.78 -0.02  0.05 -0.09 0.51 0.49 1.0
## impsafe  0.15  0.09  0.51  0.03 0.31 0.69 1.2
## impdiff  0.21  0.30 -0.25  0.28 0.42 0.58 3.8
## ipfrule  0.08  0.01  0.54 -0.04 0.30 0.70 1.1
## ipudrst -0.10  0.50  0.09  0.02 0.29 0.71 1.1
## ipmodst -0.39  0.10  0.47  0.08 0.40 0.60 2.1
## ipgdtim -0.14 -0.03  0.07  0.85 0.55 0.45 1.1
## impfree  0.09  0.06  0.07  0.38 0.21 0.79 1.3
## iphlppl -0.02  0.63  0.06 -0.05 0.41 0.59 1.0
## ipsuces  0.74  0.04  0.06  0.03 0.58 0.42 1.0
## ipstrgv  0.10  0.24  0.39  0.01 0.28 0.72 1.8
## ipadvnt  0.28  0.05 -0.32  0.39 0.56 0.44 2.8
## ipbhprp  0.01  0.24  0.53 -0.01 0.43 0.57 1.4
## iprspot  0.63 -0.21  0.33 -0.01 0.39 0.61 1.8
## iplylfr -0.08  0.53  0.11  0.13 0.36 0.64 1.3
## impenv  -0.06  0.58  0.04 -0.07 0.33 0.67 1.1
## imptrad  0.10  0.24  0.22 -0.07 0.15 0.85 2.5
## impfun   0.17 -0.18 -0.04  0.63 0.55 0.45 1.3
##
##                        ML3  ML2  ML4  ML1
## SS loadings           2.57 1.96 1.87 1.67
## Proportion Var        0.12 0.09 0.09 0.08
## Cumulative Var        0.12 0.22 0.30 0.38
## Proportion Explained  0.32 0.24 0.23 0.21
## Cumulative Proportion 0.32 0.56 0.79 1.00
##
##  With factor correlations of
##       ML3  ML2   ML4   ML1
## ML3  1.00 0.18 -0.13  0.66
## ML2  0.18 1.00  0.35  0.27
## ML4 -0.13 0.35  1.00 -0.25
## ML1  0.66 0.27 -0.25  1.00
##
## Mean item complexity =  1.7
## Test of the hypothesis that 4 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 132  and the objective function was  0.38
##
## The root mean square of the residuals (RMSR) is  0.03
```

```
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  552.55
with prob <  8e-53
## The total n.obs was  1795  with Likelihood Chi Square =  669.79  wi
th prob <  7.1e-73
##
## Tucker Lewis Index of factoring reliability =  0.911
## RMSEA index =  0.048  and the 90 % confidence intervals are  0.044
0.051
## BIC =  -319.26
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML3  ML2  ML4 ML
1
## Correlation of (regression) scores with factors   0.92 0.87 0.87 0.
9
## Multiple R square of scores with factors          0.85 0.76 0.75 0.
8
## Minimum correlation of possible factor scores     0.69 0.52 0.51 0.
6
```

```
fa.diagram(fa3, cut = 0.4)
```




```
print(fa3$loadings,cutoff = 0.4)
```

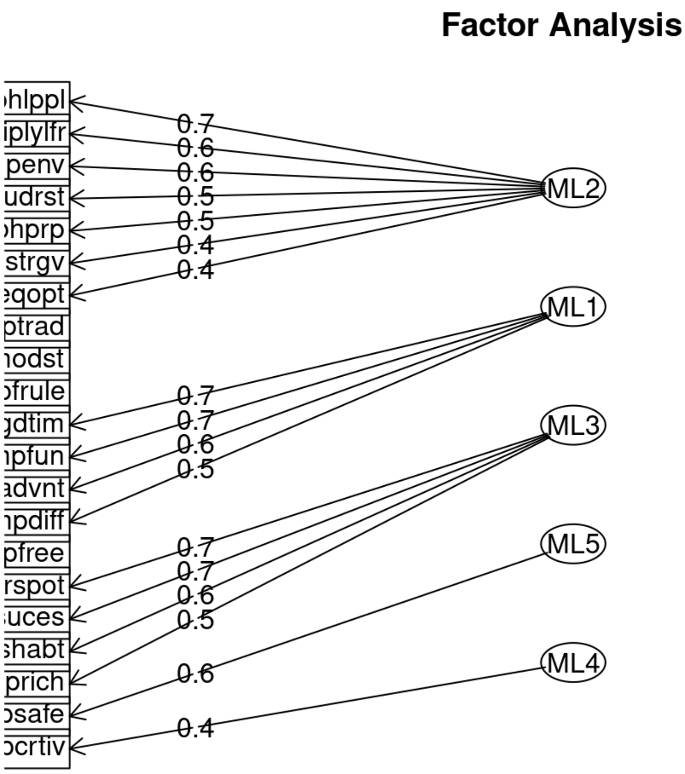
```
##
## Loadings:
##         ML3    ML2    ML4    ML1
## ipcrtiv  0.481
## imprich  0.574
## ipeqopt
## ipshabt  0.782
## impsafe                0.509
## impdiff
## ipfrule                0.544
## ipudrst         0.503
## ipmodst                0.468
## ipgdtim                       0.850
## impfree
## iphlppl         0.629
## ipsuces  0.739
## ipstrgv
## ipadvnt
## ipbhprp                0.530
## iprspot  0.630
## iplylfr         0.528
## impenv          0.577
## imptrad
## impfun                        0.629
##
##                  ML3   ML2   ML4   ML1
## SS loadings    2.514 1.866 1.741 1.566
## Proportion Var 0.120 0.089 0.083 0.075
## Cumulative Var 0.120 0.209 0.292 0.366
```

```
fa4 = fa(est, nfactors = 5,fm = 'ml', rotate = 'varimax')
fa4
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 5, rotate = "varimax", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML2   ML1   ML3   ML5   ML4   h2   u2 com
## ipcrtiv  0.15  0.31  0.32 -0.03  0.43 0.41 0.59 3.1
## imprich -0.14  0.33  0.51  0.17  0.11 0.43 0.57 2.3
## ipeqopt  0.40  0.05  0.00  0.30  0.00 0.26 0.74 1.9
## ipshabt  0.00  0.26  0.61  0.12  0.26 0.52 0.48 1.9
## impsafe  0.28  0.00  0.12  0.62 -0.01 0.48 0.52 1.5
## impdiff  0.14  0.52  0.14  0.00  0.41 0.47 0.53 2.2
## ipfrule  0.32 -0.15  0.15  0.28 -0.26 0.29 0.71 3.9
## ipudrst  0.53  0.05 -0.07  0.07  0.05 0.29 0.71 1.1
## ipmodst  0.36 -0.21 -0.21  0.23 -0.36 0.40 0.60 4.0
## ipgdtim  0.07  0.70  0.09  0.02  0.01 0.51 0.49 1.1
## impfree  0.11  0.39  0.16  0.10  0.10 0.21 0.79 1.8
## iphlppl  0.65  0.04  0.00 -0.01  0.07 0.43 0.57 1.0
## ipsuces  0.11  0.35  0.66 -0.02  0.18 0.60 0.40 1.8
## ipstrgv  0.42  0.00  0.13  0.28 -0.08 0.28 0.72 2.0
## ipadvnt -0.07  0.61  0.29 -0.27  0.23 0.59 0.41 2.2
## ipbhprp  0.52 -0.11  0.09  0.31 -0.20 0.42 0.58 2.1
## iprspot  0.04  0.12  0.66  0.05 -0.14 0.47 0.53 1.2
## iplylfr  0.57  0.16 -0.02  0.06  0.03 0.36 0.64 1.2
## impenv   0.56  0.01 -0.07  0.05  0.08 0.33 0.67 1.1
## imptrad  0.37 -0.04  0.13  0.07 -0.10 0.17 0.83 1.5
## impfun  -0.13  0.68  0.30 -0.10  0.01 0.58 0.42 1.5
##
##                        ML2  ML1  ML3  ML5  ML4
## SS loadings           2.52 2.27 1.95 0.94 0.81
## Proportion Var        0.12 0.11 0.09 0.04 0.04
## Cumulative Var        0.12 0.23 0.32 0.37 0.40
## Proportion Explained  0.30 0.27 0.23 0.11 0.10
## Cumulative Proportion 0.30 0.56 0.79 0.90 1.00
##
## Mean item complexity =  1.9
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 115  and the objective function was  0.28
##
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  401.11
with prob <  2.5e-33
## The total n.obs was  1795  with Likelihood Chi Square =  498.6  wit
h prob <  3.4e-49
##
```

```
## Tucker Lewis Index of factoring reliability =  0.927
## RMSEA index =  0.043  and the 90 % confidence intervals are  0.039
0.047
## BIC =  -363.07
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML2  ML1  ML3  M
L5   ML4
## Correlation of (regression) scores with factors   0.87 0.85 0.84 0.
72  0.70
## Multiple R square of scores with factors          0.76 0.72 0.71 0.
53  0.48
## Minimum correlation of possible factor scores     0.52 0.45 0.42 0.
05 -0.03
```

```
fa.diagram(fa4, cut = 0.4)
```




```
print(fa4$loadings,cutoff = 0.4)
```

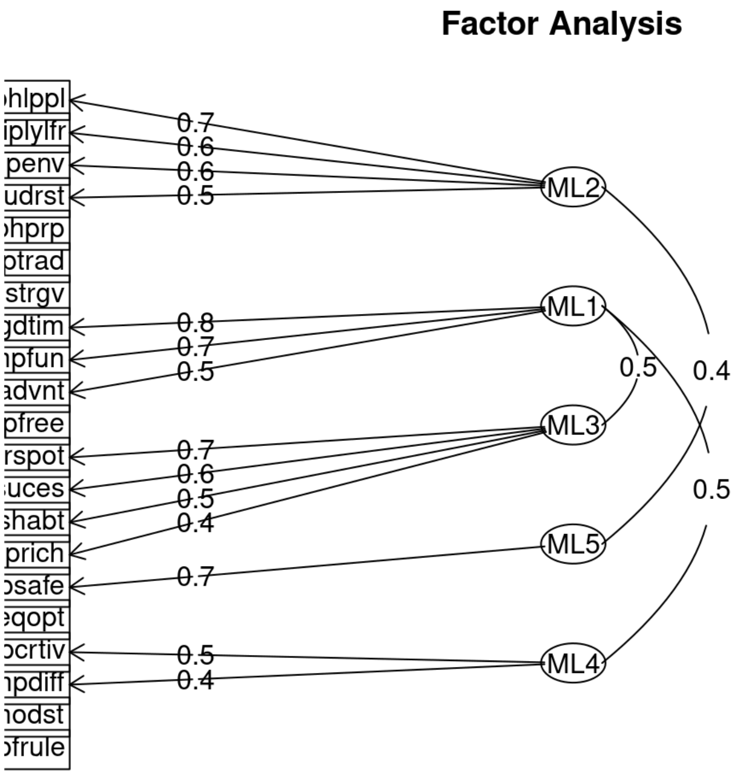
```
##
## Loadings:
##         ML2    ML1    ML3    ML5    ML4
## ipcrtiv                              0.428
## imprich                0.507
## ipeqopt  0.405
## ipshabt                0.608
## impsafe                       0.623
## impdiff         0.517                0.412
## ipfrule
## ipudrst  0.528
## ipmodst
## ipgdtim         0.702
## impfree
## iphlppl  0.651
## ipsuces                0.655
## ipstrgv  0.421
## ipadvnt         0.609
## ipbhprp  0.517
## iprspot                0.660
## iplylfr  0.574
## impenv   0.561
## imptrad
## impfun          0.681
##
##                  ML2   ML1   ML3   ML5   ML4
## SS loadings    2.518 2.268 1.948 0.937 0.813
## Proportion Var 0.120 0.108 0.093 0.045 0.039
## Cumulative Var 0.120 0.228 0.321 0.365 0.404
```

```
fa5 = fa(est, nfactors = 5, fm = 'ml',rotate = 'oblimin')
fa5
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 5, rotate = "oblimin", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML2   ML1   ML3   ML5   ML4   h2   u2 com
## ipcrtiv  0.18  0.06  0.24  0.00  0.46 0.41 0.59 1.9
## imprich -0.23  0.20  0.41  0.21  0.13 0.43 0.57 3.0
## ipeqopt  0.29  0.05 -0.04  0.32  0.01 0.26 0.74 2.1
## ipshabt -0.06  0.02  0.55  0.14  0.29 0.52 0.48 1.7
## impsafe  0.02  0.00  0.03  0.69  0.04 0.48 0.52 1.0
## impdiff  0.15  0.35 -0.03  0.06  0.44 0.47 0.53 2.2
## ipfrule  0.19 -0.10  0.21  0.25 -0.28 0.29 0.71 4.0
## ipudrst  0.52  0.02 -0.06  0.05  0.03 0.29 0.71 1.1
## ipmodst  0.26 -0.03 -0.15  0.20 -0.39 0.40 0.60 2.7
## ipgdtim  0.05  0.76 -0.08  0.08 -0.03 0.51 0.49 1.1
## impfree  0.07  0.35  0.05  0.14  0.09 0.21 0.79 1.6
## iphlppl  0.68 -0.03  0.04 -0.05  0.04 0.43 0.57 1.0
## ipsuces  0.11  0.12  0.63 -0.04  0.16 0.60 0.40 1.3
## ipstrgv  0.30 -0.01  0.13  0.27 -0.09 0.28 0.72 2.6
## ipadvnt  0.05  0.49  0.18 -0.26  0.19 0.59 0.41 2.2
## ipbhprp  0.39 -0.07  0.13  0.28 -0.22 0.42 0.58 2.9
## iprspot -0.01  0.02  0.71  0.00 -0.18 0.47 0.53 1.1
## iplylfr  0.56  0.14 -0.03  0.05  0.01 0.36 0.64 1.1
## impenv   0.56 -0.04 -0.05  0.03  0.07 0.33 0.67 1.1
## imptrad  0.34 -0.05  0.19  0.02 -0.13 0.17 0.83 1.9
## impfun  -0.10  0.68  0.15 -0.07 -0.04 0.58 0.42 1.2
##
##                        ML2  ML1  ML3  ML5  ML4
## SS loadings           2.22 1.95 1.89 1.22 1.21
## Proportion Var        0.11 0.09 0.09 0.06 0.06
## Cumulative Var        0.11 0.20 0.29 0.35 0.40
## Proportion Explained  0.26 0.23 0.22 0.14 0.14
## Cumulative Proportion 0.26 0.49 0.71 0.86 1.00
##
##  With factor correlations of
##       ML2   ML1  ML3   ML5   ML4
## ML2  1.00  0.03 0.02  0.43 -0.03
## ML1  0.03  1.00 0.46 -0.08  0.49
## ML3  0.02  0.46 1.00  0.17  0.29
## ML5  0.43 -0.08 0.17  1.00 -0.18
## ML4 -0.03  0.49 0.29 -0.18  1.00
##
## Mean item complexity =  1.8
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 115  and the objective function was  0.28
##
```

```
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  401.11
with prob <  2.5e-33
## The total n.obs was  1795  with Likelihood Chi Square =  498.6  wit
h prob <  3.4e-49
##
## Tucker Lewis Index of factoring reliability =  0.927
## RMSEA index =  0.043  and the 90 % confidence intervals are  0.039
0.047
## BIC =  -363.07
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML2  ML1  ML3  M
L5  ML4
## Correlation of (regression) scores with factors   0.88 0.89 0.89 0.
83 0.82
## Multiple R square of scores with factors          0.78 0.80 0.78 0.
68 0.67
## Minimum correlation of possible factor scores     0.55 0.60 0.57 0.
36 0.33
```

```
fa.diagram(fa5, cut = 0.4)
```



```
print(fa5$loadings,cutoff = 0.4)
```

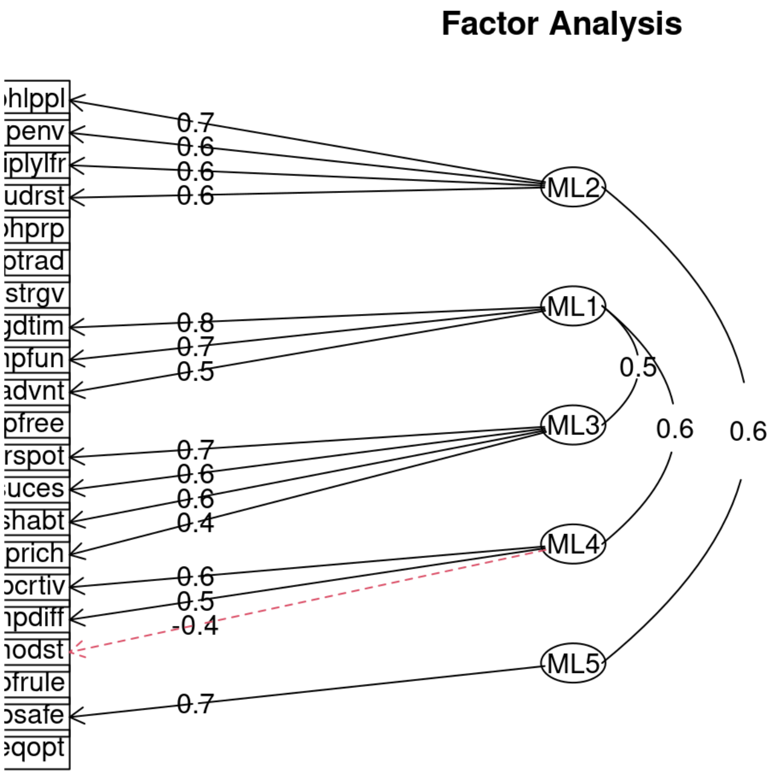
```
##
## Loadings:
##         ML2    ML1    ML3    ML5    ML4
## ipcrtiv                              0.460
## imprich               0.413
## ipeqopt
## ipshabt               0.549
## impsafe                      0.687
## impdiff                              0.440
## ipfrule
## ipudrst  0.517
## ipmodst
## ipgdtim         0.760
## impfree
## iphlppl  0.676
## ipsuces               0.629
## ipstrgv
## ipadvnt         0.489
## ipbhprp
## iprspot               0.712
## iplylfr  0.562
## impenv   0.562
## imptrad
## impfun          0.683
##
##                  ML2   ML1   ML3   ML5   ML4
## SS loadings    2.048 1.634 1.647 0.998 0.927
## Proportion Var 0.098 0.078 0.078 0.048 0.044
## Cumulative Var 0.098 0.175 0.254 0.301 0.345
```

```
fa6 = fa(est, nfactors = 5, fm = 'ml',rotate = 'promax')
fa6
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est, nfactors = 5, rotate = "promax", fm = "ml")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML2   ML1   ML3   ML4   ML5   h2   u2 com
## ipcrtiv  0.20 -0.01  0.23  0.55  0.03 0.41 0.59 1.6
## imprich -0.27  0.18  0.43  0.14  0.22 0.43 0.57 2.9
## ipeqopt  0.28  0.05 -0.08  0.03  0.31 0.26 0.74 2.2
## ipshabt -0.06 -0.05  0.56  0.37  0.15 0.52 0.48 2.0
## impsafe -0.04 -0.02  0.00  0.05  0.73 0.48 0.52 1.0
## impdiff  0.16  0.30 -0.04  0.48  0.10 0.47 0.53 2.1
## ipfrule  0.18 -0.07  0.19 -0.28  0.18 0.29 0.71 3.6
## ipudrst  0.56  0.02 -0.11  0.08  0.00 0.29 0.71 1.1
## ipmodst  0.26  0.04 -0.18 -0.43  0.12 0.40 0.60 2.3
## ipgdtim  0.02  0.80 -0.09 -0.10  0.02 0.51 0.49 1.1
## impfree  0.06  0.35  0.04  0.09  0.12 0.21 0.79 1.5
## iphlppl  0.74 -0.03 -0.01  0.12 -0.13 0.43 0.57 1.1
## ipsuces  0.14  0.09  0.63  0.24 -0.09 0.60 0.40 1.5
## ipstrgv  0.31  0.00  0.10 -0.06  0.22 0.28 0.72 2.1
## ipadvnt  0.06  0.48  0.19  0.20 -0.30 0.59 0.41 2.5
## ipbhprp  0.40 -0.05  0.09 -0.19  0.20 0.42 0.58 2.2
## iprspot  0.00  0.03  0.73 -0.15 -0.11 0.47 0.53 1.1
## iplylfr  0.60  0.15 -0.07  0.04 -0.02 0.36 0.64 1.2
## impenv   0.61 -0.05 -0.09  0.13 -0.02 0.33 0.67 1.2
## imptrad  0.37 -0.04  0.16 -0.09 -0.05 0.17 0.83 1.6
## impfun  -0.12  0.72  0.17 -0.09 -0.14 0.58 0.42 1.3
##
##                        ML2  ML1  ML3  ML4  ML5
## SS loadings           2.36 1.87 1.86 1.34 1.05
## Proportion Var        0.11 0.09 0.09 0.06 0.05
## Cumulative Var        0.11 0.20 0.29 0.35 0.40
## Proportion Explained  0.28 0.22 0.22 0.16 0.12
## Cumulative Proportion 0.28 0.50 0.72 0.88 1.00
##
##  With factor correlations of
##       ML2  ML1  ML3   ML4   ML5
## ML2  1.00 0.06 0.13 -0.21  0.60
## ML1  0.06 1.00 0.45  0.59  0.04
## ML3  0.13 0.45 1.00  0.20  0.32
## ML4 -0.21 0.59 0.20  1.00 -0.30
## ML5  0.60 0.04 0.32 -0.30  1.00
##
## Mean item complexity =  1.8
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model =  210  with the objective function =  5.53 with Chi
Square =  9872.69
## df of  the model are 115  and the objective function was  0.28
##
```

```
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  401.11
with prob <  2.5e-33
## The total n.obs was  1795  with Likelihood Chi Square =  498.6  wit
h prob <  3.4e-49
##
## Tucker Lewis Index of factoring reliability =  0.927
## RMSEA index =  0.043  and the 90 % confidence intervals are  0.039
0.047
## BIC =  -363.07
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML2  ML1  ML3  M
L4  ML5
## Correlation of (regression) scores with factors   0.89 0.90 0.88 0.
85 0.84
## Multiple R square of scores with factors          0.80 0.81 0.78 0.
72 0.71
## Minimum correlation of possible factor scores     0.60 0.62 0.56 0.
45 0.41
```

```
fa.diagram(fa6, cut = 0.4)
```



```
print(fa6$loadings,cutoff = 0.4)
```

```
##
## Loadings:
##          ML2    ML1    ML3    ML4    ML5
## ipcrtiv                       0.551
## imprich                0.429
## ipeqopt
## ipshabt                0.556
## impsafe                              0.731
## impdiff                       0.481
## ipfrule
## ipudrst  0.556
## ipmodst                      -0.431
## ipgdtim         0.803
## impfree
## iphlppl  0.741
## ipsuces                0.632
## ipstrgv
## ipadvnt         0.482
## ipbhprp
## iprspot                0.725
## iplylfr  0.602
## impenv   0.610
## imptrad
## impfun          0.716
##
##                  ML2   ML1   ML3   ML4   ML5
## SS loadings    2.354 1.683 1.694 1.192 1.010
## Proportion Var 0.112 0.080 0.081 0.057 0.048
## Cumulative Var 0.112 0.192 0.273 0.330 0.378
```

Models with 5 factors are slightly better statistically than models with 4 factors (TLI is higher (0.927 > 0.911), RMSEA is lower (0.043 < 0.048) and BIC is also lower (-363 < -319)). In general all the metrics are okay for each model. The problem with 5 factor models is that they have weak factors with 1-2 variables with each rotation method. I choose the fa2 with 4 factors and oblimin rotation method. Models with 4 factors are better interpreted. Factors division aligns with the Schwartz theory of basic human values. Model with oblimin rotation is the best choice because it has no cross-loadings and factors consist of at least 3 variables with mostly similar loadings. Moreover, chosen model shows that openness to change (ML1) and self-enhancement (ML3) are positively correlated (0.53), as well as self-transcendence (ML2) and conservation (ML4) (0.28), which again aligns with Schwartz theory (Schwartz, S. H., 1992). Below is the interpretation of factors:

ML1 - Openness to Change

- *Important to have a good time*
- *Important to seek fun and things that give pleasure*

- *Important to seek adventures and have an exciting life*

ML2 - Self-Transcendence

- *Important to help people and care for others well-being*
- *Important to care for nature and environment*
- *Important to be loyal to friends and devote to people close*
- *Important to understand different people*

ML3 - Self-Enhancement

- *Important to show abilities and be admired*
- *Important to be successful and that people recognize achievements*
- *Important to get respect from others*
- *Important to be rich, have money and expensive things*
- *Important to think new ideas and being creative*

ML4 - Conservation

- *Important to do what is told and follow rules*
- *Important to behave properly*
- *Important to live in secure and safe surroundings*
- *Important to be humble and modest, not draw attention*

6. Making a model without some variables

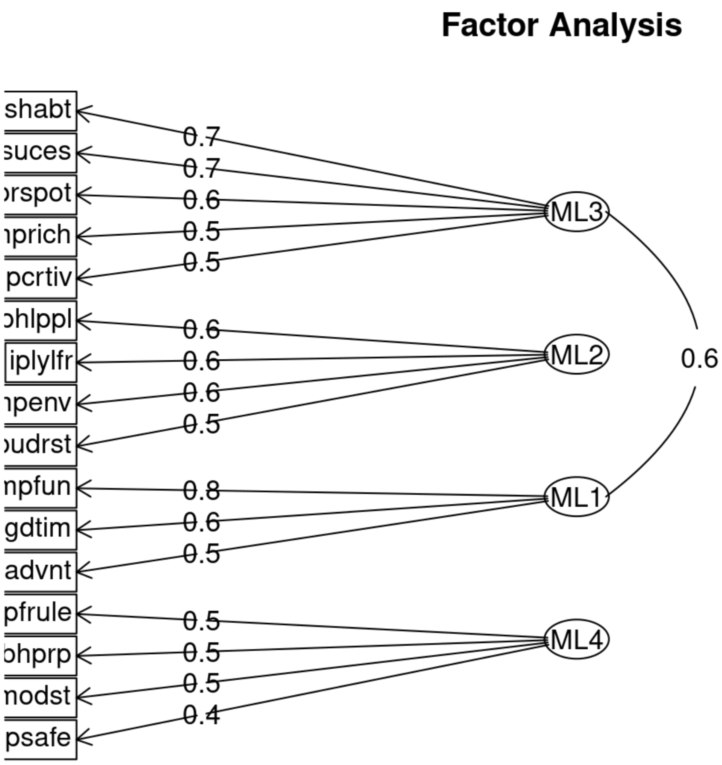
```
est_clean <- est %>%
  select(-ipeqopt, -impdiff, -impfree, -ipstrgv, -imptrad)

fa_final = fa(est_clean, nfactors = 4, fm = 'ml',rotate = 'oblimin')
fa_final
```

```
## Factor Analysis using method =  ml
## Call: fa(r = est_clean, nfactors = 4, rotate = "oblimin", fm = "m
l")
## Standardized loadings (pattern matrix) based upon correlation matri
x
##           ML3   ML2   ML1   ML4   h2   u2 com
## ipcrtiv  0.46  0.27  0.03 -0.25 0.35 0.65 2.2
## imprich  0.52 -0.17  0.18  0.09 0.42 0.58 1.5
## ipshabt  0.71 -0.01  0.00 -0.01 0.51 0.49 1.0
## impsafe  0.21  0.12 -0.06  0.44 0.28 0.72 1.6
## ipfrule  0.12  0.05 -0.07  0.53 0.33 0.67 1.1
## ipudrst -0.05  0.50  0.02  0.11 0.30 0.70 1.1
## ipmodst -0.30  0.15 -0.01  0.50 0.42 0.58 1.9
## ipgdtim -0.03  0.14  0.64  0.03 0.40 0.60 1.1
## iphlppl  0.04  0.64 -0.04  0.04 0.43 0.57 1.0
## ipsuces  0.70  0.10  0.09 -0.03 0.60 0.40 1.1
## ipadvnt  0.19  0.09  0.50 -0.29 0.56 0.44 2.0
## ipbhprp  0.09  0.27 -0.07  0.51 0.45 0.55 1.6
## iprspot  0.58 -0.13  0.03  0.24 0.38 0.62 1.4
## iplylfr -0.02  0.58  0.12  0.09 0.38 0.62 1.1
## impenv  -0.02  0.55 -0.04  0.04 0.32 0.68 1.0
## impfun   0.02 -0.07  0.82  0.02 0.68 0.32 1.0
##
##                        ML3  ML2  ML1  ML4
## SS loadings           2.16 1.66 1.60 1.37
## Proportion Var        0.14 0.10 0.10 0.09
## Cumulative Var        0.14 0.24 0.34 0.42
## Proportion Explained  0.32 0.24 0.24 0.20
## Cumulative Proportion 0.32 0.56 0.80 1.00
##
##  With factor correlations of
##       ML3  ML2   ML1   ML4
## ML3  1.00 0.07  0.57 -0.06
## ML2  0.07 1.00  0.02  0.29
## ML1  0.57 0.02  1.00 -0.30
## ML4 -0.06 0.29 -0.30  1.00
##
## Mean item complexity =  1.4
## Test of the hypothesis that 4 factors are sufficient.
##
## df null model =  120  with the objective function =  4.14 with Chi
Square =  7405.45
## df of  the model are 62  and the objective function was  0.14
##
## The root mean square of the residuals (RMSR) is  0.02
## The df corrected root mean square of the residuals is  0.03
##
## The harmonic n.obs is  1795 with the empirical chi square  186.74
with prob <  2e-14
```

```
## The total n.obs was  1795  with Likelihood Chi Square =  244.96  wi
th prob <  1.4e-23
##
## Tucker Lewis Index of factoring reliability =  0.951
## RMSEA index =  0.041  and the 90 % confidence intervals are  0.035
0.046
## BIC =  -219.59
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##                                                    ML3  ML2  ML1  M
L4
## Correlation of (regression) scores with factors   0.90 0.85 0.90 0.
83
## Multiple R square of scores with factors          0.82 0.72 0.80 0.
69
## Minimum correlation of possible factor scores     0.63 0.45 0.61 0.
38
```

```
fa.diagram(fa_final, cut = 0.4)
```




```
print(fa_final$loadings,cutoff = 0.4)
```

```
##
## Loadings:
##         ML3    ML2    ML1    ML4
## ipcrtiv  0.456
## imprich  0.521
## ipshabt  0.714
## impsafe                       0.444
## ipfrule                       0.534
## ipudrst         0.504
## ipmodst                       0.496
## ipgdtim                0.638
## iphlppl         0.638
## ipsuces  0.698
## ipadvnt                0.496
## ipbhprp                       0.515
## iprspot  0.581
## iplylfr         0.575
## impenv          0.550
## impfun                 0.816
##
##                  ML3   ML2   ML1   ML4
## SS loadings    2.015 1.573 1.394 1.225
## Proportion Var 0.126 0.098 0.087 0.077
## Cumulative Var 0.126 0.224 0.311 0.388
```

I deleted variables whcih were not included in any factor in fa2. New model is better because it has higher TLI (now it is > 95%!) and lower RMSEA. BIC is bigger but still negative. Factors stayed the same but there are significant changes in variables loadings in ML1: ipgdtim loading decreased (0.8 to 0.6) and impfun became the dominant variable (loading incresed from 0.6 to 0.8). Now all variables are included in factors and have loadings > 0.4. Correalarions between factors remained.

7. Saving factor scores and selecting new variables

```
factor_scores <- as.data.frame(fa_final$scores)
names(factor_scores) <- c("Openness_to_Change", "Self_Transcendence",
                          "Self_Enhancement", "Conservation")

est_clean$id <- 1:nrow(est_clean)
factor_scores$id <- 1:nrow(factor_scores)

est_lg <- estonia %>%
  select(stflife, gndr, agea) %>%
  mutate(
    stflife = ifelse(stflife %in% c(77, 88, 99), NA, stflife),
    gndr = ifelse(gndr == 9, NA, gndr),
    agea = ifelse(agea == 999, NA, agea)
  ) %>%
  mutate(id = row_number()) %>%
  filter(id %in% est_clean$id) %>%
  mutate(
    across(everything(), as.numeric),
    gndr = factor(gndr,
                  levels = c(1, 2),
                  labels = c("Male", "Female"))
  ) %>%
  drop_na()

est_with_fs <- est_lg %>%
  left_join(factor_scores, by = "id") %>%
  select(-id)

str(est_with_fs)
```

```
## 'data.frame':	1792 obs. of  7 variables:
##  $ stflife           : num  7 3 7 7 4 6 10 0 7 2 ...
##  $ gndr              : Factor w/ 2 levels "Male","Female": 1 2 1 2
2 1 2 1 2 2 ...
##  $ agea              : num  30 75 38 48 62 33 23 65 34 53 ...
##  $ Openness_to_Change: num  0.454 0.283 0.642 0.922 -1.285 ...
##  $ Self_Transcendence: num  1.839 0.754 0.828 1.072 1.252 ...
##  $ Self_Enhancement  : num  -0.5877 -0.0373 0.5905 1.4732 -0.9788
...
##  $ Conservation      : num  1.861 0.0773 0.5938 1.2445 -0.6661 ...
```

I chose new variables for linear regression:

- *stflife - How satisfied with life as a whole (0 - extremely dissatisfied, 10 - extremely satisfied)*
- *agea - Age*
- *gndr - Gender (Male and Female)*

8. Linear regression

Research question: Do value orientations predict life satisfaction, and which values make the greatest contribution?

```
m1 <- lm(stflife ~ agea + gndr, data = est_with_fs)
summary(m1)
```

```
##
## Call:
## lm(formula = stflife ~ agea + gndr, data = est_with_fs)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.579 -1.345  0.133  1.607  5.361
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.87253    0.14339  47.931  < 2e-16 ***
## agea        -0.02659    0.00268  -9.922  < 2e-16 ***
## gndrFemale   0.45130    0.10640   4.241 2.33e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.196 on 1789 degrees of freedom
## Multiple R-squared:  0.05682,    Adjusted R-squared:  0.05576
## F-statistic: 53.89 on 2 and 1789 DF,  p-value: < 2.2e-16
```

```
m2 <- lm(stflife ~ agea + gndr + Openness_to_Change, data = est_with_f
s)
summary(m2)
```

```
##
## Call:
## lm(formula = stflife ~ agea + gndr + Openness_to_Change, data = est
_with_fs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.6277 -1.3407  0.1199  1.6006  5.3397
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          6.87132    0.14337  47.928  < 2e-16 ***
## agea                -0.02654    0.00268  -9.903  < 2e-16 ***
## gndrFemale           0.44915    0.10640   4.221 2.55e-05 ***
## Openness_to_Change  0.06982    0.05747   1.215    0.225
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.196 on 1788 degrees of freedom
## Multiple R-squared:  0.0576, Adjusted R-squared:  0.05601
## F-statistic: 36.42 on 3 and 1788 DF,  p-value: < 2.2e-16
```

```
m3 <- lm(stflife ~ agea + gndr + Openness_to_Change + Self_Transcenden
ce, data = est_with_fs)
summary(m3)
```

```
##
## Call:
## lm(formula = stflife ~ agea + gndr + Openness_to_Change + Self_Tran
scendence,
##     data = est_with_fs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.5928 -1.3616  0.1207  1.5701  5.5235
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         6.867338   0.143196  47.958  < 2e-16 ***
## agea               -0.026476   0.002677  -9.892  < 2e-16 ***
## gndrFemale          0.450903   0.106270   4.243 2.32e-05 ***
## Openness_to_Change  0.083443   0.057691   1.446   0.1483
## Self_Transcendence -0.144262   0.061149  -2.359   0.0184 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.193 on 1787 degrees of freedom
## Multiple R-squared:  0.06052,    Adjusted R-squared:  0.05842
## F-statistic: 28.78 on 4 and 1787 DF,  p-value: < 2.2e-16
```

```
m4 <- lm(stflife ~ agea + gndr + Openness_to_Change + Self_Transcenden
ce + Self_Enhancement, data = est_with_fs)
summary(m4)
```

```
##
## Call:
## lm(formula = stflife ~ agea + gndr + Openness_to_Change + Self_Tran
scendence +
##     Self_Enhancement, data = est_with_fs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.3875 -1.3713  0.1251  1.5909  5.5372
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         6.867648   0.142946  48.044  < 2e-16 ***
## agea               -0.026553   0.002672  -9.937  < 2e-16 ***
## gndrFemale          0.456475   0.106105   4.302 1.78e-05 ***
## Openness_to_Change  0.232520   0.079839   2.912  0.00363 **
## Self_Transcendence -0.155300   0.061179  -2.538  0.01122 *
## Self_Enhancement   -0.215693   0.080004  -2.696  0.00708 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.189 on 1786 degrees of freedom
## Multiple R-squared:  0.06433,    Adjusted R-squared:  0.06171
## F-statistic: 24.56 on 5 and 1786 DF,  p-value: < 2.2e-16
```

```
m5 <- lm(stflife ~ agea + gndr + Openness_to_Change + Self_Transcenden
ce + Self_Enhancement + Conservation, data = est_with_fs)
summary(m5)
```

```
##
## Call:
## lm(formula = stflife ~ agea + gndr + Openness_to_Change + Self_Tran
scendence +
##     Self_Enhancement + Conservation, data = est_with_fs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.3828 -1.3783  0.1264  1.5902  5.5309
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)         6.867902   0.143005  48.025  < 2e-16 ***
## agea               -0.026557   0.002673  -9.935  < 2e-16 ***
## gndrFemale          0.456436   0.106135   4.301  1.8e-05 ***
## Openness_to_Change  0.234598   0.082221   2.853  0.00438 **
## Self_Transcendence -0.151905   0.069041  -2.200  0.02792 *
## Self_Enhancement   -0.220260   0.090842  -2.425  0.01542 *
## Conservation       -0.008398   0.079062  -0.106  0.91542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.19 on 1785 degrees of freedom
## Multiple R-squared:  0.06434,    Adjusted R-squared:  0.06119
## F-statistic: 20.46 on 6 and 1785 DF,  p-value: < 2.2e-16
```

```
anova(m1, m2, m3, m4, m5)
```

```
## Analysis of Variance Table
##
## Model 1: stflife ~ agea + gndr
## Model 2: stflife ~ agea + gndr + Openness_to_Change
## Model 3: stflife ~ agea + gndr + Openness_to_Change + Self_Transcen
dence
## Model 4: stflife ~ agea + gndr + Openness_to_Change + Self_Transcen
dence +
##     Self_Enhancement
## Model 5: stflife ~ agea + gndr + Openness_to_Change + Self_Transcen
dence +
##     Self_Enhancement + Conservation
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1   1789 8628.4
## 2   1788 8621.3  1     7.115 1.4837 0.223359
## 3   1787 8594.5  1    26.769 5.5823 0.018250 *
## 4   1786 8559.7  1    34.836 7.2646 0.007099 **
## 5   1785 8559.7  1     0.054 0.0113 0.915424
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4 model is the best with the least RSS. Adding Conservation does not improve model. I will interpret model 4. Age has negative correlation with life satisfaction. Females have higher life satisfaction than males. Openness to change leads to higher life satisfaction, self-transcendence and self-enhancement have negative correlation with life satisfaction. Value orientation do predict life satisfaction and openness to change has the biggest effect among all, but in general all the effects are small.

References: Schwartz, S. H. (1992). Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries. In M. P. Zanna (Ed.), Advances in experimental social psychology (Vol. 25, pp. 1–65). Academic Press.